

.1 GOAL

To decoke furnace tubes of B101, B102, B103, and B1001 and B151 preheater as per guide **GM-07-000-00-001** and per the Montreal Quality Control Manual requirement.

- B-101 : decoking radiant and convection tubes combined, two circuits (North coil and South coil)
- B-102 : decoking radiant and convection tubes, two different circuits (portion of convection coil and complete radiant coil together and upper two rows of convection)
- B-103: : decoking radiant and convection tubes combined, one circuit
- B-1001 : decoking radiant and convection tubes combined, two different circuits (Central convection coils and upper radiant coils combined and outer convection coils and lower radiant coils combined)
- B-151: decoking preheater, one circuit

.2 SCHEDULES

Execution date of the work: April 2026 on two 10-hour shifts

The contractor must provide a detailed schedule per furnace according to the two options listed in pt #3

.3 TENDER FORM IN TWO OPTIONS

- OPTION 1 : One furnace after another
 - o Using 1 decoke truck + 1 filter press
- OPTION 2 : Two furnaces in parallel (and the 5th furnace after)
 - o Using 2 decoke trucks + 2 filter press

.4 Scope of work - supplied by contractor

The Contractor must provide the labor, supervision, materials, machinery and equipment required to do, but not be limited to, the decoking of the tubes (convection and radiant) of furnaces mentioned above, by following the details cited in the guide GM-07-000-00-001.

See list of equipment to be supplied by the Contractor:

- *Foam witness pigs to determine the diameter of the first cleaning pig*
- *Cleaning pigs of different diameters suited for furnace tubes. The use of pigs with tungsten carbide tips is*

PROHIBITED

- *The final pigs of adequate size to confirm the cleanliness of the tubes*
- *Swab pigs to adequately dry furnace tubes*
- *Pumper truck(s), filter press and hoses required for decoking*
- *Launchers / receivers for the 4 furnaces and 1 preheater with in particular having for the B-102 launchers and receivers with vents or drains of 2 inch for decoking the radiant section so that we can connect a nitrogen line to dry the tubes.*
- *Flexible hoses and rigid pipes and fittings for decoking, also ball valves at the exit of the truck. (see # 6 for details)*
- *3inch portable gas pumps + hoses.*
- *Provide basins under the filter-press and under the connections in back of pumper truck(s) to recover the soda ash which is repeatedly spilled*
- *Make an initial hydraulic test to assess the condition of the tubes and final after the end of the decoking work*

.5 Documents required by contractor

The Contractor must issue the following:

- *Schedules as described in pt # 2*
- *Submission slips as described in point # 3*
- *Description of the equipment used for decoking*
- *Detail of labor expected*
- *The list of hourly rates for labor, equipment, pensions, transportation, materials. etc.*
- *A sketch for each furnace showing the launchers / receivers and the various equipment including the lengths of the hoses required VS truck locations and the required connections to the launchers and receivers and to the truck*

Special conditions for furnace B-101:

- *Decoking in 2 circuits:*
 - o *Circuit # 1: radiant and convection combined (north side)*
 - o *Circuit # 2: radiant and convection combined (south side)*
- *Launchers / receivers required for:*

- o 4 " - 300 # (qty: 2)
- o 10 " - 150 # (qty:2)
- Provide various pigs of diameters for 10 ", 8 ", 6 ", 5 ", 4 " tubes.

Particular conditions on furnace B-102 :

- Decoking in 2 circuits :
 - o Circuit #1 : Upper two rows of convection piped together
 - o Circuit #2 : Lower 8 rows of convection coils and complete radiant coils piped together

*****SODA ASH**

- Launchers / receivers required for :
 - o 4"-300# (qty :2) upper convection coils
 - o 8"-300# (qty:2) lower convection and radiant coils
- DECOKING WITH SODA ASH AND TUBE DRYING WITH NITROGEN for radiant only
- Provide a filter press with transfer pump
- With the drying pig (SWAB), flush with nitrogen at the end to get the water out, then disconnect the hose from the truck and vent it in a safe place. See guide GM-07-000- 00-001 for more details (to be coordinated with Suncor operation / execution)

Special conditions for furnace B-103:

- Decoking in 1 circuit
- Launchers / receivers required for:
 - o 4 " - 300 # (qty: 2)
- Provide 4 " pigs

Special conditions for furnace B-151:

- Decoking in 1 circuit
- Launchers / receivers required for:
 - o 4 " - 300 # (qty: 2)
- Provide 4 " pigs

Special conditions for furnace B-1001:

- Decoking in 2 circuits:
 - o Circuit # 1: convection and radiant together (radiant bottom section)
 - o Circuit # 2: convection and radiant together (radiant top section)

- *Launchers and receivers required*
 - o 4 " - 300 # (qty: 4)
- *Provide 4 " pigs*
- *The launchers and receivers are 5 feet from the ground next to the FVs (see PID in the appendix)*

.6 Technical data of furnaces,

	B-102	B-101	B-1001
REQUIRED LAUNCHERS AND RECEIVERS	4"-300# (qty :2) for convection 8"-300# (qty.2) for radiant	4"-300# (qty :2) for convection 10"-150# (qty.2) for radiant	4"-300# (qty :4)
TUBE METALURGY (CONVECTION)	A335 gr P5 (5CR-1/2Mo)	A335 gr P5 (5CR-1/2Mo)	SA213 T5 (5CR-1/2Mo)
TUBE DIAMETER AND SCH CONVECTION	4" sch 40	4" SCH 40	OD 4.5" THK. 0.207"
TUBE METALURGY (RADIANT)	A335 gr P5 et P9 (9CR -1Mo) 5Cr 1/2Mo SS 347 H	A 335 GR P9 (9 Cr – 1 Mo)	SA213 T9 (9CR-1Mo)
TUBE DIAMETER AND SCH RADIANT	6"- 8" sch 40	MOSTLY 4" SCH 40 10" sch 40, 8" sch 40 6" sch 40, 5" sch40	OD 4.5" THK. 0.207"
MAXIMUM PRESSURE THAT CAN BE USED FOR DECOKING AND CLEANING	740 PSIG	740 PSIG	740 PSIG
AMBIANT TEMPERATURE DURING DECOKING	100 F	100 F	100 F
TUBE LENGHT CONVECTION	± 1000 FT (4")	± 1600 FT (4")	± 2050 FT (4")
TUBES LENGHT RADIANT	± 5000 FT (6"-8")	4" (±1800 FT) FROM BOTTOM 5" (±360 FT) FROM	± 4600 FT (4")

		BOTTOM 6" (±100 FT) 8" (±100 FT) 10" (±100 FT) 4" (±500 FT) ON TOP	
TOTALE TUBE LENGHT FOR DECOKING (CONVECTION AND RADIANT)	± 6000 FT	± 4600 FT	± 6650 FT
TOTAL VOLUME OF SODA-ASH SOLUTION (GAL)	21000 GAL (2 COILS IN RADIANT (6" ET 8") + LAUNCHERS (2_ + HOSE + TRUCK	N/A NO SS IN TUBES	N/A NO SSIN TUBES
TUBE POSITION	HORIZONTAL	HORIZONTAL	HORIZONTAL
EXCEPTION POSITION	N/A	N/A	VERTICAL ON 12 FT FOR 4"
TUBE HEIGHT CONVECTION	34 FT	33 FT	43 FT
TUBE HEIGHT RADIANT	5 FT	25 FT	13 FT

	B-103	B-151
REQUIRED LAUNCHERS AND RECEIVERS	4"-300# (qty :2)	4"-300# (qty :2)
TUBE METALURGY (CONVECTION)	A335 gr P5 (5CR-1/2Mo)	A106 gr B
TUBE DIAMETER AND SCH CONVECTION	4" sch 40	4" sch 40
TUBE METALURGY (RADIANT)	A335 gr P5 (5CR-1/2Mo)	N/A
TUBE DIAMETER AND SCH RADIANT	4" sch 40	N/A
MAXIMUM PRESSURE THAT CAN BE USED FOR DECOKING AND CLEANING	740 PSIG	740 PSIG
AMBIANT TEMPERATURE DURING DECOKING	100 F	100 F
TUBE LENGHT CONVECTION	± 600 FT (4")	± 3400 FT (4")
TUBES LENGHT RADIANT	± 1000 FT (4")	N/A
TOTALE TUBE LENGHT FOR DECOKING (CONVECTION AND RADIANT)	± 1600 FT	± 3400 FT
TOTAL VOLUME OF SODA- ASH SOLUTION (GAL)	N/A	N/A No SS IN TUBES

TUBE POSITION	VERTICAL	HORIZONTAL
EXCEPTION POSITION	HORIZONTAL ON 24 FT FOR 4''	N/A
TUBE HEIGHT CONVECTION	5 FT	5 FT
TUBE HEIGHT RADIANT	35 FT	N/A

.7 Scope of work – supplied by Suncor

Those directly involved in the decoking operation must read the GM-07-000-00-001 guide and this directive.

SUNCOR must provide the workforce, supervision, materials, machinery and equipment required to enable the decoking operation, among others see below:

Requirements for the 4 furnaces and 1 pre-heater

- i. Provide safety training to the Contractor's personnel
- ii. Provide monitoring of the progress of the decoking operation
- iii. Install the scaffolding
- iv. Obtain and ensure zero energy
- v. Remove all pipes and instruments and install temporary spools in their places
- vi. Provide water including hoses and fittings (hydrant)
- vii. Provide a min 375 scfm air compressor including hoses and fittings
- viii. Provide containment for coke and other products
- ix. Provide electricity and lighting
- x. Provide diesel for anti-foam
- xi. Provide labor and tools for handling and installing hoses and launchers / receivers
- xii. Cleaning facilities for the truck tank
- xiv. Fuel for the decoking truck pumps as well as for the 3inch portable pumps.

Special conditions for furnace B-102:

- i. Suncor will supply a “baker tank” filled with the 21000 gal of new soda ash solution (follow GM-07... for the recipe and quantity)*
- ii. Suncor will provide a 2nd baker tank for used soda ash*
- iii. We will have a vacuum truck on hand during decoking to recover the liquids in all basins.*
- iv. We will provide 150 psig nitrogen (supplied by Suncor) including hoses and fittings (was used in 2018 and 2021)*
- v. Suncor will provide basins under the launchers / receivers for the radiant section to recover the soda ash which is repeatedly spilled + have the availability of a vacuum truck during decoking to recover the liquids in all basins*

APPENDIX “A”***English portion of the GM-07-000-00-001 – Furnace tube pigging*****APPENDIX “B”*****Plot plan of the units*****APPENDIX “C”*****B-101 Annotated PIDs and various drawings*****APPENDIX “D”*****B-102 Annotated PIDs and various drawings*****APPENDIX “E”*****B-1001 Annotated PIDs and various drawings*****APPENDIX “F”*****B-103 Annotated PIDs and various drawings*****APPENDIX “G”*****B-151 Annotated PIDs and various drawings***